

MCNP simulation of the effects of prostatic edema in LDR brachytherapy with ¹²⁵I using the adapted ICRP 110 voxel male phantom

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1. INTRODUCTION

Prostate cancer is the second most common cancer among men in Brazil, especially after the age of 60. It is estimated that between 2023 and 2025, there will be approximately 71,730 new cases per year, with almost 17,000 deaths recorded in 2023. One of the treatments used is LDR brachytherapy, which inserts ¹²⁵I seeds into the prostate. This procedure can cause edema, altering the volume of the gland and the position of the seeds, which affects the distribution of radiation. This study evaluated this impact by simulating increases in prostate volume, from 10% to 50%, in the ICRP 110 male phantom, using an automated system to adjust the position of the seeds in simulations performed with the MCNP code.

2. MATERIALS AND METHODS

2.1 REX phantom (Voxel Model, ICRP 110)

The REX phantom, recommended by ICRP 110 and used in this study, is a male voxel model based on tomography images of a reference adult. Composed of 220 slices and 34.6 mm³ voxels, it includes 122 segmented structures, such as organs, tissues, and bones, according to ICRP anatomical data.

2.2 Radioactive Isotope ¹²⁵I

In this work, the ¹²⁵I Amersham model 6711 source was used as a reference. It is modeled as an inner silver cylinder containing an emitting film. The outer shell, made of titanium, has a diameter of 0.81 mm and a length of 4.5 mm, and is closed by two ellipsoids.

2.3 Using the developed system

The system's "MCNP Input Calculation/Generation" option creates the input file based on the prostate volume, automatically calculating the required number of seeds. The user defines the seed positions based on data provided by the Moritz Geometry Tool software, which offers a 3D visualization of the anatomy. The system also simulates edema ranging from 10% to 50%, automatically adjusting the seed distribution and prostate volume in the phantom after loading it into the proposed scenario, with automatic modifications to the surrounding voxel, lattice, and parallelepiped planes. The analysis is restricted to the prostate region. The graphical interface was designed to reflect the clinical model, using columns A–M and rows 0–12, where the user defines the positions, quantity, and depth of the seeds (Figure 1). Finally, the system generates the corresponding instructions to compose the input file.

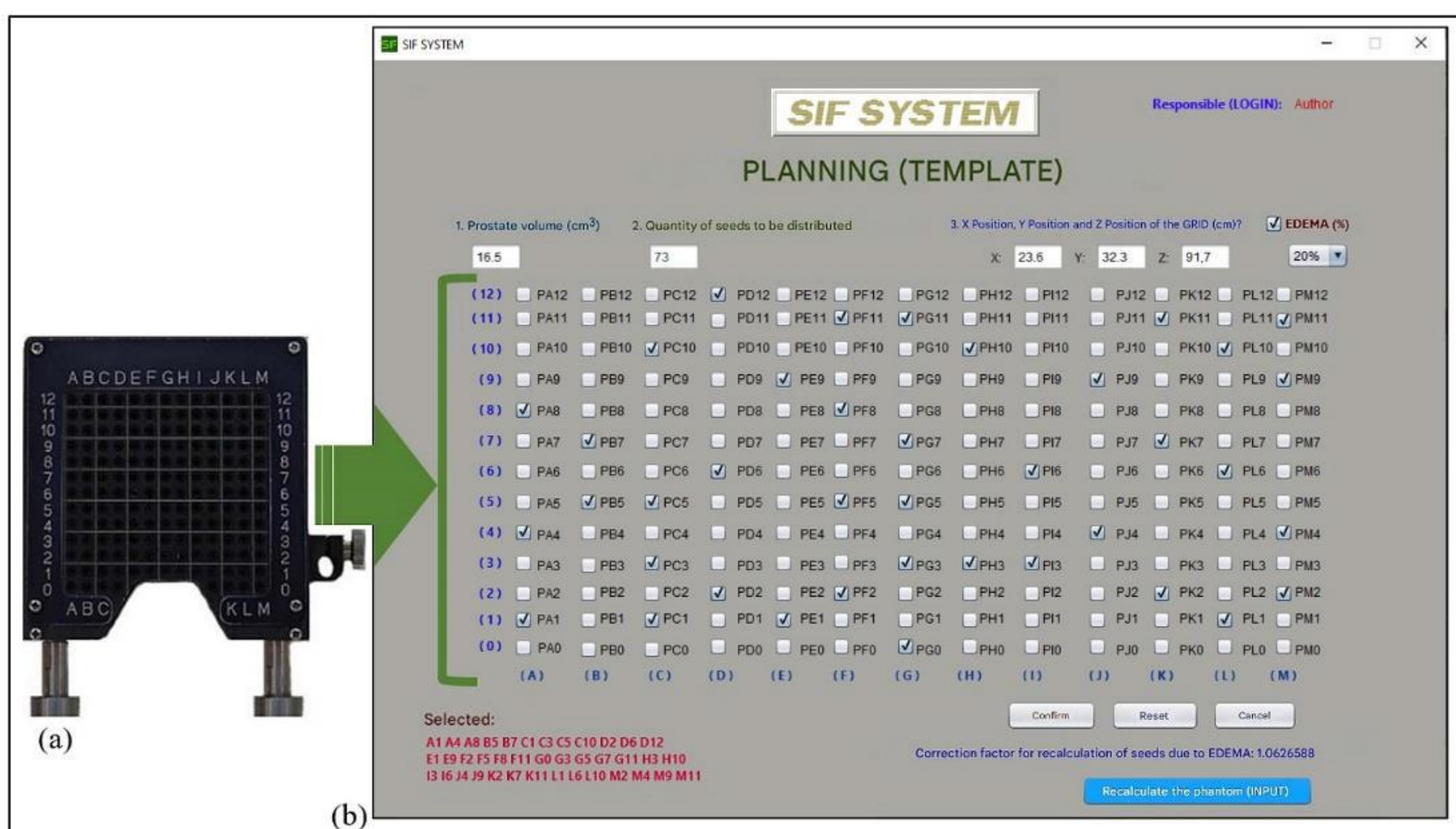


Figure 1: (a) The template used in the clinical procedure. (b) The computer system and the correlation with the template.

3. RESULTS

The results of the computer simulation provide a better understanding of how dose distribution is affected in low dose rate (LDR) brachytherapy for prostate cancer, especially in relation to prostatic edema. A progressive reduction in the dose delivered to the prostate was observed as the volume increased from 10% to 50% (Table 1). This difference can compromise the effectiveness of treatment if not considered in planning. To analyze this relationship, dose-volume histograms (DVHs) were generated, which showed a decrease in coverage and dose homogeneity as edema progresses (Figure 2). Since edema is transient and decreases over time, its evolution needs to be taken into account during planning. Simulations provide valuable data for the clinical team to adjust treatment strategies, ensuring better tumor coverage and protection of sensitive tissues, even with variations in prostate volume.

Table 1 - Dose Values and Gap Percentages for Conditions with and without Prostatic Edema.

EDEMA (%)	Dose to organ (Gy) <i>Planned (144 Gy)</i>	Percentage of total Gy planned (%)
0	144.00	100
10	142.56	99
20	139.68	97
30	135.54	94
40	131.04	91
50	125.28	87

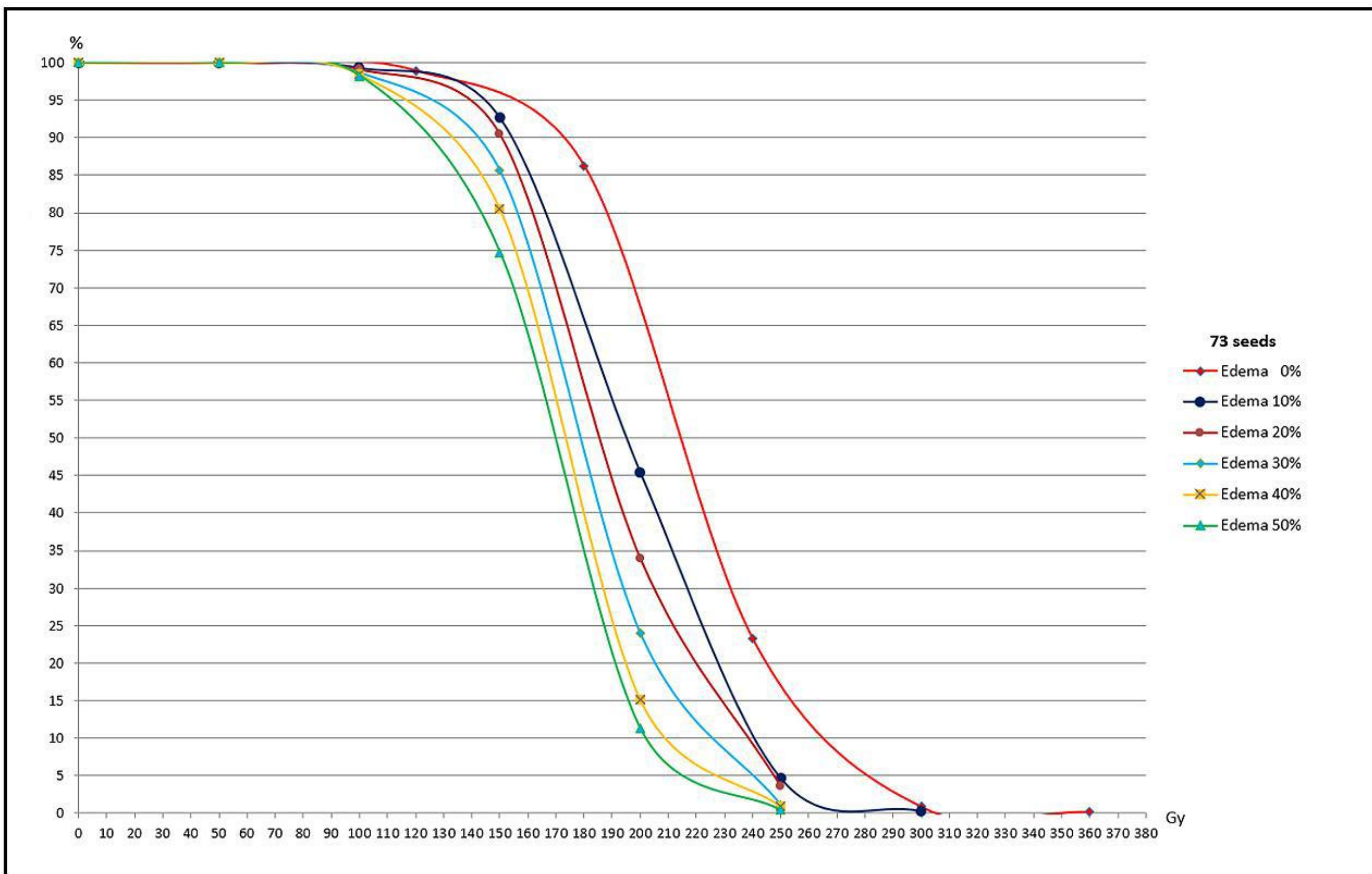


Figure 2: Dose-Volume Histograms (DVHs) of the results of the MCNP code input files in the EDEMA study.

4. CONCLUSIONS

This study demonstrates the importance of considering prostatic edema in low dose rate (LDR) brachytherapy for prostate cancer. Simulations with the REX phantom in the MCNPX code showed that increases in prostate volume between 30% and 50% can significantly compromise dose distribution, especially in cases of prolonged inflammation, affecting the effectiveness of treatment with ¹²⁵I seeds. Changes in implant geometry directly impact the effect between seeds and tumor coverage. As each patient has a unique anatomy, it is essential to include edema in planning and post-implantation follow-up to ensure the appropriate therapeutic dose and avoid side effects. Thus, the study reinforces the importance of integrating computer simulation, clinical knowledge, and physiological monitoring to optimize therapeutic outcomes and patients' quality of life.

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